

DiffusionSfM: Predicting Structure and Motion via Ray Origin and Endpoint Diffusion

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Project page: qitaozhao.github.io/DiffusionSfM

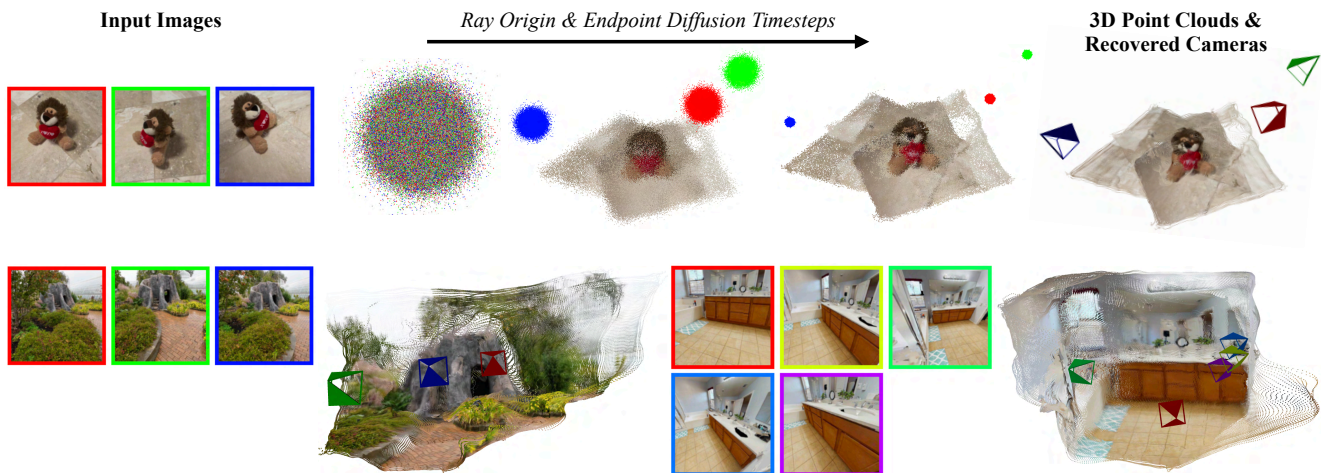


Figure 1. **DiffusionSfM.** **Top:** Given a set of multi-view images (left), DiffusionSfM represents scene geometry and cameras (right) as pixel-wise ray origins and endpoints in a global frame. It learns a denoising diffusion model to infer these elements directly from multi-view inputs. Unlike traditional Structure-from-Motion (SfM) pipelines, which separate pairwise reasoning and global optimization into two stages, our approach unifies both into a single end-to-end multi-view reasoning framework. **Bottom:** Example results of inferred scene geometry and cameras for two distinct settings: a real-world outdoor scene (left) and a synthetic indoor scene (right).

Abstract

Current Structure-from-Motion (SfM) methods typically follow a two-stage pipeline, combining learned or geometric pairwise reasoning with a subsequent global optimization step. In contrast, we propose a data-driven multi-view reasoning approach that directly infers 3D scene geometry and camera poses from multi-view images. Our framework, DiffusionSfM, parameterizes scene geometry and cameras as pixel-wise ray origins and endpoints in a global frame and employs a transformer-based denoising diffusion model to predict them from multi-view inputs. To address practical challenges in training diffusion models with missing data and unbounded scene coordinates, we introduce specialized mechanisms that ensure robust learning. We empirically validate DiffusionSfM on both synthetic and real datasets, demonstrating that it outperforms classical and learning-based approaches while naturally modeling uncertainty.

1. Introduction

The task of recovering structure (geometry) and motion (cameras) from multi-view images has long been a focus of the computer vision community, with typical pipelines [25] performing pairwise correspondence estimation followed by global optimization. While classical methods relied on hand-designed features, matching, and optimization, there has been a recent shift towards incorporating learning-based alternatives [5, 6, 14, 23]. More recently, the widely influential DUST3R [36] advocates for predicting pairwise 3D pointmaps (instead of only correspondences), demonstrating that this can yield accurate dense geometry and cameras. In order to reconstruct more than two views, DUST3R (and its variants [10]) still require a global optimization reminiscent of classic bundle adjustment. While these methods, both classical and learning-based, have led to impressive improvements in SfM, the overall approach is largely unchanged – learned or geometric pairwise reasoning followed by global optimization. In this work, we

seek to develop an alternative approach that directly predicts both structure and motion, unifying pairwise reasoning and global optimization into a single multi-view framework.

We are of course not the first to attempt to find unified alternatives to the two-stage SfM pipeline. In the sparse-view setting where conventional correspondence-based methods struggle, several works employ multi-view architectures to jointly reason across input images. SparsePose [26], RelPose++ [12], and PoseDiffusion [34] all leverage multi-view transformers to estimate camera pose for input images, albeit using differing mechanisms such as regression, energy-based modeling, and denoising diffusion. More recently, RayDiffusion [41] argues for a local raymap parameterization of cameras instead of a global extrinsic matrix and shows that existing patch-based transformers can be easily adapted for this task, yielding significantly more accurate pose predictions. Importantly, such methods predict only camera motion and fail to predict scene structure.

In this work, we present DiffusionSfM, an end-to-end multi-view model that directly infers dense 3D geometry and cameras from multiple input images. Instead of inferring (depth-agnostic) rays per image patch (as in RayDiffusion [41]) or 3D points per pixel (as in DUST3R [36]), DiffusionSfM effectively combines both to predict ray *origins and endpoints* per pixel, directly reporting both scene geometry (endpoints) and generalized cameras (rays). These can readily be converted back to traditional cameras [41]. Compared to RayDiffusion, our model directly predicts structure as well as motion at a finer scale (pixel-wise v.s. patch-wise). Compared to DUST3R, our model directly predicts motion as well as structure but, even more importantly, does so for N views, eliminating the need for memory-intensive global alignment. To model uncertainty, we train a denoising diffusion model but find two key challenges that need to be addressed. First, diffusion models require (noisy) ground truth as input for training, but existing real datasets do not have known endpoints for all pixels due to missing depth in multi-view stereo. Second, the 3D coordinates of endpoints can be potentially unbounded, whereas diffusion models require normalized data. We develop mechanisms to overcome these challenges, leveraging additional “GT mask conditioning” as input to inform the model of missing input data and parameterizing 3D points in projective space instead of Euclidean space. We find these strategies allow us to learn accurate predictions for structure and motion.

We train and evaluate DiffusionSfM on real-world and synthetic datasets [21, 24, 42] and find that it can infer accurate geometry and cameras for both object-centric and scene-level images (see Fig. 1). In particular, we find that DiffusionSfM yields more accurate camera estimates compared to prior work across these settings while also modeling the underlying uncertainty via the diffusion process. In summary, we show that DiffusionSfM can serve as a unified

multi-view reasoning model for 3D geometry and cameras.

2. Related Work

Structure from Motion. Structure-from-Motion (SfM) systems [25] aim to simultaneously recover geometry and cameras given a set of input images. The typical SfM pipeline extracts pixel correspondences from keypoint matching [2, 16], and performs global bundle adjustment (BA) to optimize sparse 3D points and camera parameters by minimizing reprojection errors. Recently, SfM pipelines have been substantially enhanced by replacing classical subcomponents with learning-based methods, such as neural feature descriptors [6, 8], keypoint matching [15, 23, 30], and bundle adjustment [13, 32].

More recently, an emerging body of research aims to unify the various SfM subcomponents into an end-to-end neural framework. Notably, ACEZero [3] fits a single neural network to input images and learns pixel-aligned 3D coordinates in a self-supervised manner, while FlowMap [27] predicts per-frame cameras and depth maps using off-the-shelf optical flow as supervision. Though ACEZero and FlowMap are promising attempts to revolutionize SfM pipelines, they both register images incrementally and may suffer under large viewpoint changes. DUST3R [36] directly regresses 3D pointmaps from image pairs and shows strong generalization ability [11, 21]. Building on DUST3R, MAST3R [10] introduces a feature head, offering pixel-matching capabilities. MAST3R-SfM [7] is a more scalable SfM pipeline based on MAST3R. While these approaches [7, 10, 36] show impressive performance and robustness under sparse views, they are essentially pair-based, requiring sophisticated global alignment procedures to form a consistent estimate for more than two views.

Pose Estimation with Global Reasoning. For the task of sparse-view pose estimation, learning-based methods equipped with global reasoning show favorable robustness where traditional SfM methods [25, 28] fail. This line of research includes energy-based [12, 40], regression-based [26], and diffusion-based pose estimators [34, 41]. Among them, diffusion-based methods show a better ability to handle uncertainty [41]. Closest to our work, RayDiffusion [41] leverages a denoising diffusion model [9, 19] with a patch-aligned (depth-unaware) ray representation to predict generic cameras. Our method goes further and pursues a generic representation for both geometry and cameras in the form of ray origins and endpoints for each pixel. In addition to resulting in a richer output, this joint geometry and pose prediction also yields improvements for pose estimation.

3. Method

Given a set of sparse (*i.e.*, 2-8) input images, DiffusionSfM predicts the geometry and cameras of a 3D scene in a global

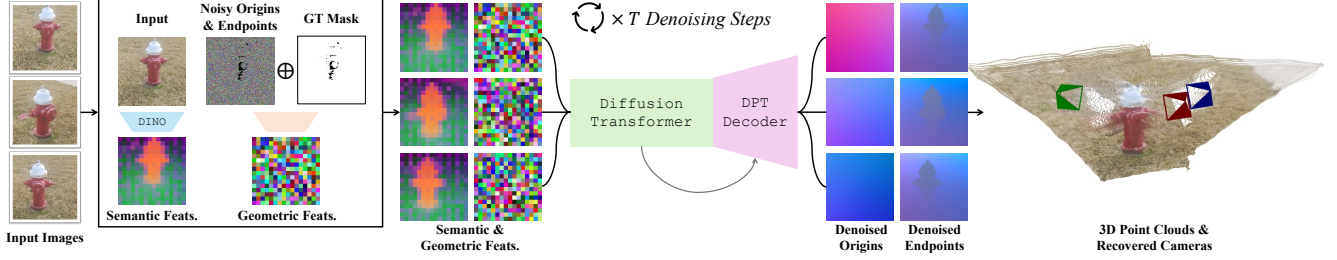


Figure 2. **Method.** Given sparse multi-view images as input, DiffusionSfM predicts pixel-wise ray origins and endpoints in a global frame (Sec. 3.1) using a denoising diffusion process (Sec. 3.2). For each image, we compute patch-wise embeddings with DINOv2 [18] and embed noisy ray origins and endpoints into latents using a single downsampling convolutional layer, ensuring alignment with the spatial footprint of the image embeddings. We implement a Diffusion Transformer architecture that predicts clean ray origins and endpoints from noisy samples. A convolutional DPT [20] head outputs full-resolution denoised ray origins and endpoints. To handle incomplete ground truth (GT) during training, we condition the model on a GT mask (Sec. 3.3). At inference, the GT mask is set to all ones, enabling the model to predict origins and endpoints for all pixels. The predicted ray origins and endpoints can be directly visualized in 3D or post-processed to recover camera extrinsics, intrinsics, and multi-view consistent depth maps.

coordinate frame. In Sec. 3.1, we propose to represent 3D scenes as dense pixel-aligned ray origins and endpoints. To predict such scene representations from sparse input images while modeling uncertainty, Sec. 3.2 proposes a denoising diffusion architecture. We then discuss some key practical challenges in training such a model in Sec. 3.3.

3.1. 3D Scenes as Ray Origins and Endpoints

Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ with a depth map $\mathbf{D} \in \mathbb{R}^{H \times W}$, camera intrinsics $\mathbf{K} \in \mathbb{R}^{3 \times 3}$, and world-to-camera extrinsics $\mathbf{T} \in \mathbb{R}^{4 \times 4}$ (equivalently, rotation $\mathbf{R} \in SO(3)$ and translation $\mathbf{t} \in \mathbb{R}^3$), each 2D image pixel $\mathbf{P}_{ij} = [u, v]$ corresponds to a ray that travels from the camera center $\mathbf{c}$ through the pixel’s projected position on the image plane, terminating at the object’s surface as specified by the depth map $\mathbf{D}$. The endpoint of the ray associated with image pixel $\mathbf{P}_{ij}$ is given by:

$$\mathbf{E}_{ij} = \mathbf{T}^{-1} h(\mathbf{D}_{ij} \cdot \mathbf{K}^{-1}[u, v, 1]^T) \quad (1)$$

where h maps the 3D point into homogeneous coordinates. The shared ray origin $\mathbf{O}_{ij}$ for all pixels is equivalent to the camera center $\mathbf{c}$, and can be computed as:

$$\mathbf{O}_{ij} = \mathbf{c} = h(-\mathbf{R}^{-1}\mathbf{t}) \quad (2)$$

In summary, we associate each image pixel with a ray origin and endpoint $\mathbf{S}_{ij} = \langle \mathbf{O}_{ij}, \mathbf{E}_{ij} \rangle$ in world coordinates, describing the location of the observing camera and the observed 3D point on the object surface. Given a bundle of ray origins and endpoints, we can easily extract the corresponding camera pose [41].

Learning Over-Parameterized Representations. We represent 3D scenes and cameras using distributed ray origins $\mathbf{O}$ and endpoints $\mathbf{E}$ rather than global alternatives such as quaternions and translation vectors. This design is inspired by RayDiffusion [41], and it facilitates the use of the

distributed deep features learned by state-of-the-art vision backbones, such as DINOv2 [18], which encode image information in a patch-wise manner. Notably, while ray origins $\mathbf{O}$ should ideally be identical across all pixels, we predict them densely alongside ray endpoints $\mathbf{E}$. This encourages ray origins to remain close within the same image, providing implicit regularization during training. Practically, predicting both ray origins and endpoints simultaneously is easy to implement using a single projection head.

3.2. DiffusionSfM

We propose a denoising Diffusion Transformer (DiT) architecture [19] that predicts ray origins and endpoints (Sec. 3.1) via a denoising diffusion process. An overview of DiffusionSfM is given in Fig. 2.

Diffusion Framework. Given pixel-aligned ray origins and endpoints $\mathcal{S} = \text{stack}(\{\mathbf{S}^{(n)}\}_{n=1}^N)$ associated with a set of N input images, we apply a forward diffusion process [9, 29] that adds time-dependent Gaussian noise to them. Let $\mathcal{S}_t$ denote the noisy ray origins and endpoints at timestep t , where $\mathcal{S}_0$ is the clean sample and $\mathcal{S}_T$ (at the final diffusion step T) approximates pure Gaussian noise. The forward diffusion process is defined as:

$$\mathcal{S}_t = \sqrt{\bar{\alpha}_t} \mathcal{S}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad (3)$$

where $t \sim \text{Uniform}(0, T]$, $\epsilon \sim \mathcal{N}(0, I)$, and $\bar{\alpha}_t$ follows a pre-defined noise schedule that controls the strength of added noise at each timestep. To perform the reverse diffusion process, which progressively reconstructs the clean sample given noisy observations, we train a diffusion model f_θ that takes $\mathcal{S}_t$ as input and optionally incorporates additional conditioning information $\mathcal{C}$. The model is trained using the following loss function (with the “ x_0 ” objective):

$$\mathcal{L}_{\text{Diffusion}} = \mathbb{E}_{t, \mathcal{S}_0, \epsilon} \|\mathcal{S}_0 - f_\theta(\mathcal{S}_t, t, \mathcal{C})\|^2 \quad (4)$$

Architecture. We implement f_θ using a DiT [19] architecture conditioned on deep image features $\mathcal{C} \in \mathbb{R}^{N \times h \times w \times c_1}$ from DINOv2 [18], where h and w are the patch resolution and c_1 is the embedding dimension. To align pixels to the spatial information learned by DINOv2, we apply a convolutional layer with kernel size and stride equal to ViT patch size: this spatially downsamples the noisy ray origins and endpoints $\mathcal{S}_t$ to match the DINOv2 features while increasing their feature dimension:

$$\mathcal{F} = \text{Conv}(\mathcal{S}_t) \in \mathbb{R}^{N \times h \times w \times c_2} \quad (5)$$

The combined DiT input is constructed by concatenating these two feature sets along the channel dimension: $\mathcal{F} \oplus \mathcal{C}$. Within DiT, patch-wise features attend to others through self-attention [33]. To distinguish between different images and their respective patches, we apply sinusoidal positional encoding [33] based on image and patch indices.

While the DiT operates on low-resolution features, our objective is to produce pixel-aligned dense ray origins and endpoints. To achieve this, we employ a DPT (Dense Prediction Transformer) [20] decoder, which takes intermediate feature maps from both DINOv2 and DiT as inputs. The DPT decoder progressively increases the feature resolution through several convolutional layers. The final ray origins and endpoints are decoded from the DPT output using a single linear layer. During inference, we apply the trained model in the reverse diffusion process to iteratively denoise a randomly initialized Gaussian sample.

3.3. Practical Training Considerations

Homogeneous Coordinates for Unbounded Geometry. Real-world scenes often exhibit significant variations in scale, both across different scenes and within a single scene. While normalizing by the average distance of 3D points [36] helps address cross-scene variation, within-scene variations remain. For instance, a background building may be much farther away than foreground elements, potentially resulting in extremely large coordinate values when generating ground-truth ray origins and endpoints. However, neural networks (especially diffusion models) tend to train most effectively when working with bounded inputs and outputs (e.g., between -1 and 1). To stabilize training across the large scale variations present in 3D scene datasets, we propose to represent ray origins and endpoints in homogeneous coordinates. Specifically, given any 3D point, we apply a homogeneous transform from $\mathbb{R}^3 \rightarrow \mathbb{P}^3$:

$$(x, y, z) \rightarrow \frac{1}{w}(x, y, z, 1) \quad (6)$$

where w is an arbitrary scale factor. To encourage bounded coordinates in practice, we choose w such that the homogeneous coordinate is unit-norm:

$$w := \sqrt{x^2 + y^2 + z^2 + 1} \quad (7)$$

Unit normalization allows homogeneous coordinates to serve as a bounded representation for unbounded scene geometry. For example, $(x, y, z, 0)$ is a point at infinity in the direction of (x, y, z) . We find that this representation makes large coordinate values more tractable during training.

Training with Incomplete Ground Truth. Many real-world datasets (e.g., CO3D [21] and MegaDepth [11]), provide only sparse point clouds, leading to incomplete depth information. This presents a significant challenge in diffusion training, as ground-truth (GT) depth values that are used to create clean ray endpoints often contain invalid or missing data. It is highly undesirable for these missing ray endpoints to be interpreted as part of the target distribution. Unlike regression models (e.g., DUST3R [36]), which do not require (noisy) GT data as *input* and can simply apply a mask on the loss for supervision, diffusion models must handle incomplete inputs during training.

To mitigate this issue, we further apply GT masks $\mathcal{M} \in \mathbb{R}^{N \times H \times W}$ to the DiT inputs, where zero values indicate pixels with invalid depth. During training, we multiply DiT inputs with GT masks element-wise, then concatenate along the channel dimension: $\mathcal{S}'_t = (\mathcal{M} \cdot \mathcal{S}_t) \oplus \mathcal{M}$. Then, we only compute the diffusion loss in Eq. 4 over unmasked pixels. By implementing these strategies, we encourage the model to focus on regions with valid GT values during training. During inference, however, we would like the diffusion process to estimate ray endpoints at all pixels, so we always use GT masks with values set to one.

Sparse-to-Dense Training. In practice, we find that training the entire model from scratch leads to slow convergence and suboptimal performance. To address this, we propose a sparse-to-dense training approach. First, we train a sparse version of the model, where the DPT decoder is removed, and the output ray origins and endpoints have the same spatial resolution as the DINOv2 features. Unlike Eq. 5, no spatial downsampling is required, so this sparse model uses a single linear layer to embed the noisy ray origins and endpoints. Once the sparse model is trained, we initialize the dense model DiT with the learned weights from the sparse model. This two-stage approach significantly improves performance; see the supplementary material for comparisons.

4. Experiments

4.1. Experimental Setup

Datasets. We introduce two model variants, each trained on different datasets. (1) DiffusionSfM-CO3D: Following prior work [12, 41], we train and evaluate our model on the CO3D dataset [21], which consists of turntable video sequences of various object categories. Specifically, we train on 41 object categories and evaluate on both these *seen* categories and an additional 10 *unseen* categories to assess generalization. (2) DiffusionSfM: This variant is trained on the

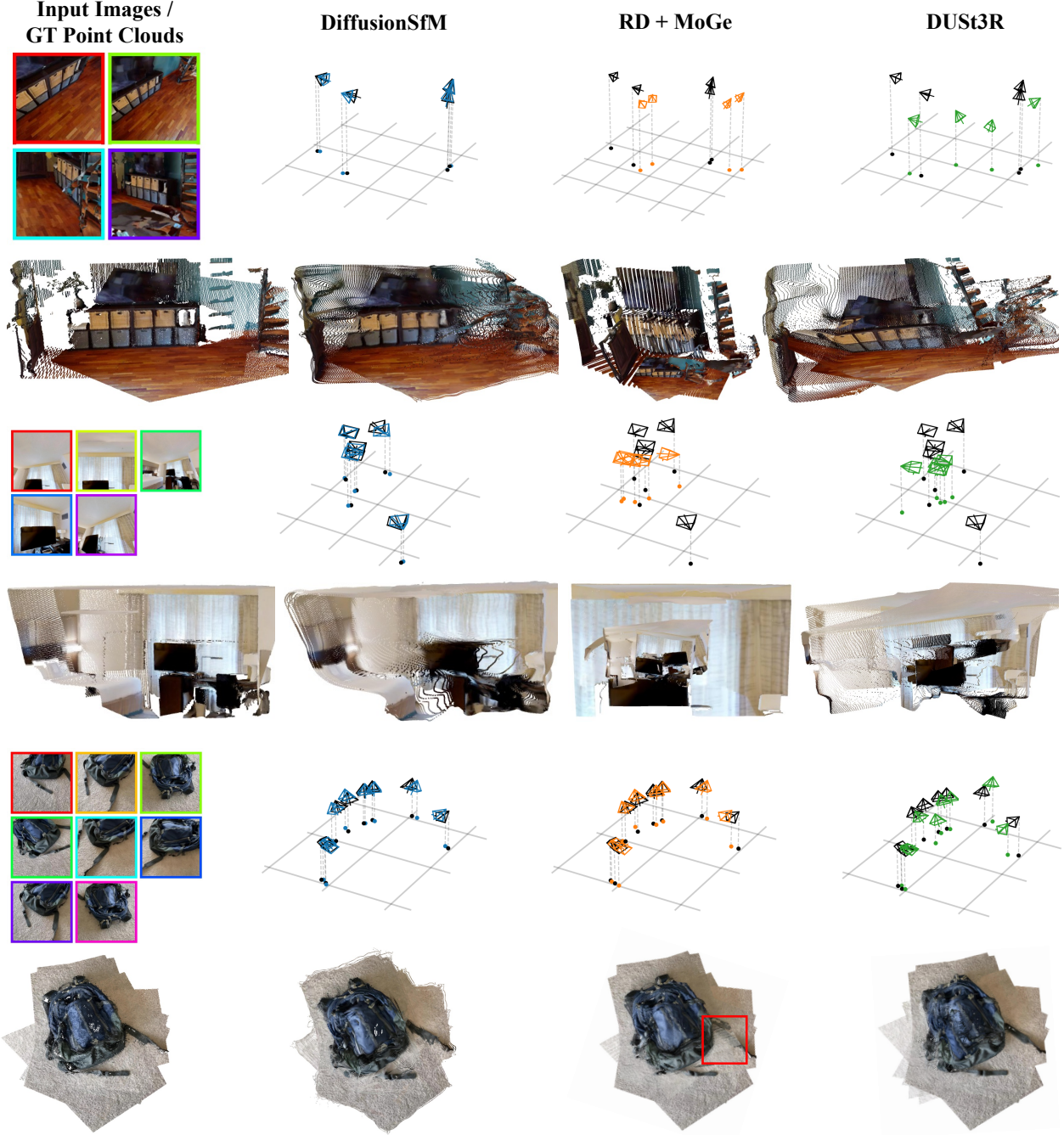


Figure 3. **Qualitative Comparison on Camera Pose Accuracy and Predicted Geometry.** For each method, we plot the ground-truth cameras in black and the predicted cameras in other colors. DiffusionSfM demonstrates robust performance even with challenging inputs. Compared to DUST3R, which sometimes fails to register images in a consistent manner, DiffusionSfM consistently yields a coherent global prediction. Additionally, while we observe that DUST3R can predict highly precise camera rotations, it often struggles with camera centers (see the backpack example). Input images depicting scenes are out-of-distribution for RayDiffusion, as it is trained on CO3D only.

datasets used for DUST3R [36], excluding Waymo [31] due to its excessively sparse depth maps. The included datasets are Habitat [24], CO3D [21], ScanNet++ [39], ArkitScenes [1], Static Scenes 3D [17], MegaDepth [11], and Blended-MVS [38]. We follow the DUST3R official repository [37]

guidelines to extract image pairs, ensuring reasonable overlap between paired images. To construct multi-view data given these pre-computed pairs, we iteratively sample images using an adjacency matrix, maintaining sufficient overlap with the selected set. Beyond CO3D, this model variant

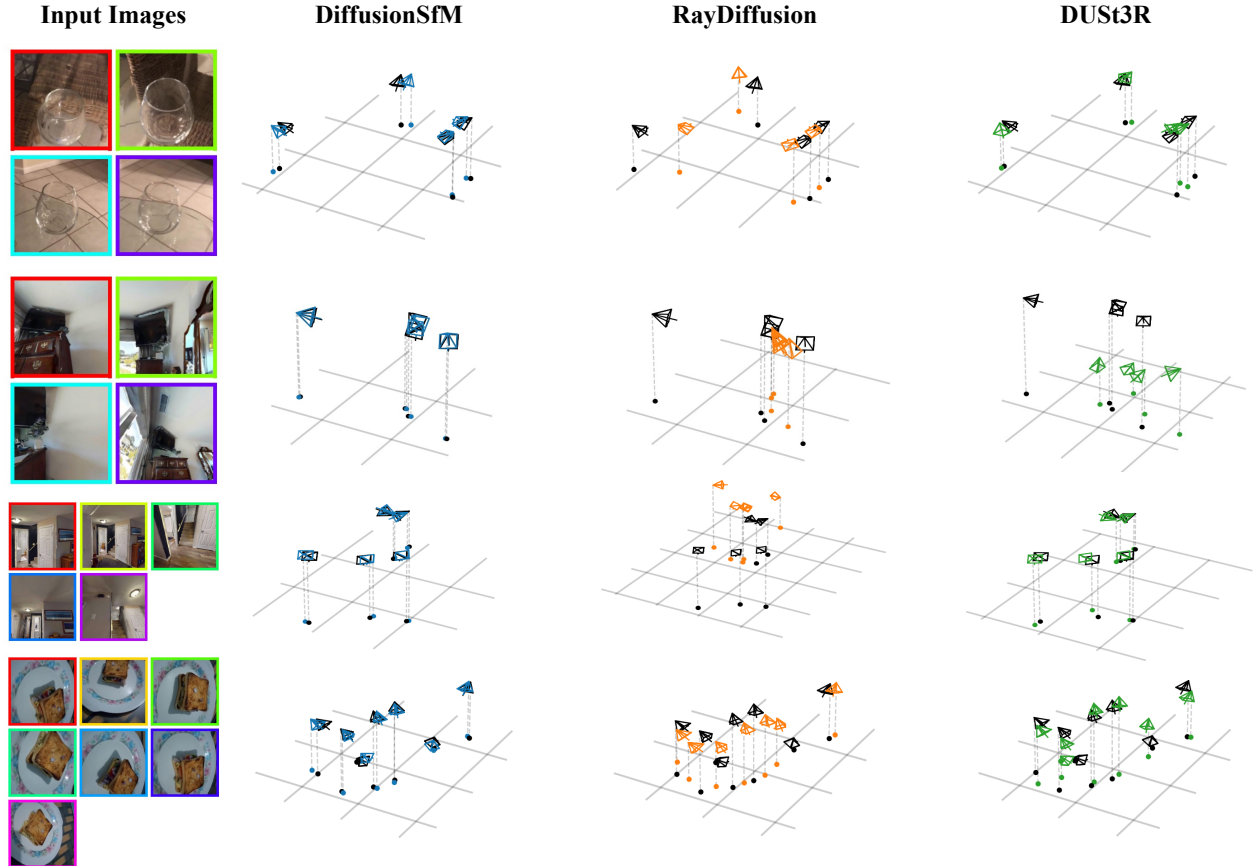


Figure 4. **Additional Qualitative Results on Predicted Camera Poses.** DiffusionSfM shows robustness to ambiguous patterns in inputs.

is also evaluated on Habitat and RealEstate10k [42].

Baselines and Metrics. To evaluate camera pose accuracy in the sparse-view setup, we compare with RayDiffusion [41] and DUST3R [36], along with previous methods [12, 22, 34]. DUST3R is trained initially on a mixture of eight datasets, while most other baselines are trained only on CO3D. For a fair and comprehensive comparison, we re-train DUST3R on the 41-10 split of CO3D, using the authors’ official implementation [37] and hyperparameters (referred to as DUST3R-CO3D). To evaluate camera predictions, we follow prior work [41] and convert predicted rays back to traditional extrinsic matrices and report two pose accuracy metrics: (1) Camera Rotation Accuracy which compares the predicted relative camera rotation between images against ground truth and (2) Camera Center Accuracy which compares predicted camera centers to the ground truth after a similarity alignment. To evaluate the estimated geometry (*i.e.*, ray endpoints), we report Chamfer Distance (CD). Additionally, to explore whether combining an off-the-shelf sparse-view pose estimation method (RayDiffusion) with a monocular depth estimation model (MoGe [35]) is sufficient to infer 3D scene geometry from

multiple images, we introduce a new baseline, RD+MoGe. We include more details in the supplementary. Unless otherwise specified, all evaluations use 2–8 input images.

4.2. Evaluation on CO3D

Camera Pose Accuracy. We present the quantitative results in Tab. 1. For camera rotation accuracy, both versions of DiffusionSfM outperform all baselines, except that DUST3R achieves slightly higher overall accuracy than DiffusionSfM-CO3D on unseen categories – likely due to its training on more data. For camera center accuracy, our approach consistently outperforms all other methods. We hypothesize that these gains stem from our explicit modeling of camera centers through ray origin prediction. The qualitative results in Fig. 3 illustrate that DiffusionSfM produces robust predictions given challenging inputs, whereas DUST3R produces inaccurate results. We attribute this improvement to our model’s probabilistic modeling capability derived from diffusion, as well as its multi-view reasoning abilities, which together effectively handle these challenging scenarios. While we observe that DUST3R can predict highly precise camera rotations, it often struggles with cam-

# of Images	Rotation Accuracy ($\uparrow$, @ 15°)								Center Accuracy ($\uparrow$, @ 0.1)							
	2	3	4	5	6	7	8		2	3	4	5	6	7	8	
Seen Categories	COLMAP [22]	30.7	28.4	26.5	26.8	27.0	28.1	30.6	100	34.5	23.8	18.9	15.6	14.5	15.0	
	PoseDiffusion [34]	75.7	76.4	76.8	77.4	78.0	78.7	78.8	100	77.5	69.7	65.9	63.7	62.8	61.9	
	RelPose++ [12]	81.8	82.8	84.1	84.7	84.9	85.3	85.5	100	85.0	78.0	74.2	71.9	70.3	68.8	
	RayDiffusion [41]	91.8	92.4	92.6	92.9	93.1	93.3	93.3	100	94.2	90.5	87.8	86.2	85.0	84.1	
	DUST3R-CO3D [36]	86.7	87.9	88.0	88.2	88.6	88.8	88.9	100	92.0	86.8	83.8	82.0	81.1	80.4	
	DUST3R [36]	91.7	92.7	93.3	93.6	93.8	94.0	94.3	100	93.0	85.7	81.9	79.6	77.8	76.8	
	DiffusionSfM-CO3D	93.4	94.0	94.5	94.8	95.0	95.2	<u>95.1</u>	100	95.9	93.6	92.2	91.2	<u>90.7</u>	90.2	
	DiffusionSfM	<u>92.4</u>	<u>93.5</u>	<u>94.0</u>	<u>94.5</u>	<u>94.7</u>	<u>95.1</u>	95.2	100	<u>95.3</u>	<u>92.9</u>	<u>91.9</u>	91.2	90.8	<u>90.1</u>	
Unseen Categories	COLMAP [22]	34.5	31.8	31.0	31.7	32.7	35.0	38.5	100	36.0	25.5	20.0	17.9	17.6	19.1	
	PoseDiffusion [34]	63.2	64.2	64.2	65.7	66.2	67.0	67.7	100	63.6	50.5	45.7	43.0	41.2	39.9	
	RelPose++ [12]	69.8	71.1	71.9	72.8	73.8	74.4	74.9	100	70.6	58.8	53.4	50.4	47.8	46.6	
	RayDiffusion [41]	83.5	85.6	86.3	86.9	87.2	87.5	88.1	100	87.7	81.1	77.0	74.1	72.4	71.4	
	DUST3R-CO3D [36]	79.8	81.5	82.6	82.7	83.0	83.3	83.7	100	83.6	77.2	71.8	70.0	68.1	67.0	
	DUST3R [36]	90.8	92.6	93.6	93.6	93.8	93.6	93.4	100	87.9	79.8	74.3	71.7	69.4	67.8	
	DiffusionSfM-CO3D	90.4	91.2	<u>92.7</u>	93.0	93.1	93.3	<u>93.5</u>	100	<u>91.1</u>	<u>87.7</u>	<u>85.3</u>	<u>83.7</u>	<u>82.7</u>	<u>82.0</u>	
	DiffusionSfM	91.5	92.8	93.6	93.9	94.1	94.2	94.4	100	92.4	87.9	86.2	85.3	84.1	83.4	

Table 1. **Camera Rotation and Center Accuracy on CO3D.** On the left, we report the proportion of relative camera rotations within 15° of the ground truth. On the right, we report the proportion of camera centers within 10% of the scene scale, relative to the ground truth. To align the predicted camera centers to ground truth, we apply an optimal similarity transform (s , $\mathbf{R}$, $\mathbf{t}$), hence the alignment is perfect at $N = 2$ but worsens with more images. DiffusionSfM outperforms all other methods for camera center accuracy, and outperforms all methods trained on equivalent data for rotation accuracy.

era centers (see the backpack example). Additional qualitative results on camera accuracy are provided in Fig. 4.

Predicted Geometry. To evaluate predicted geometry, we compute Chamfer Distance (CD) and show comparisons against baselines in Tab. 2. We compute CD in two setups (with and without foreground object mask), and find that DiffusionSfM-CO3D performs best without foreground masking. In this setup, the predicted ray endpoints corresponding to the background image pixels tend to have larger coordinate values than foreground ones, and therefore dominate CD. This result indicates that our model provides more accurate predictions for complex image backgrounds. In terms of CD with masking, DUST3R achieves the best result, while our two model variants outperform DUST3R-CO3D. See Fig. 3 for visualization. We also notice that naively combining RayDiffusion poses with estimated depths from MoGe does not work well, as the inferred depths are potentially inconsistent across different views, thus resulting in degraded performance.

4.3. Evaluation on Scene-Level Datasets

Beyond the object-centric CO3D dataset [21], we compare DiffusionSfM against DUST3R on two scene-level datasets: Habitat (in-distribution) [24] and RealEstate10k (out-of-distribution) [42]. The results are presented in Tab. 3. While DiffusionSfM achieves comparable camera rotation accuracy to DUST3R, it consistently predicts more accurate camera centers, aligning with our findings on CO3D.

# of Images	2	3	4	5	6	7	8
RD*+MoGe [41]	0.059	0.064	0.071	0.062	0.063	0.061	0.061
DUST3R* [36]	0.036	0.037	0.040	0.040	0.037	0.036	0.039
DUST3R [36]	<u>0.021</u>	<u>0.023</u>	<u>0.024</u>	<u>0.024</u>	<u>0.025</u>	<u>0.025</u>	<u>0.023</u>
DiffusionSfM*	0.019	0.021	0.022	0.022	0.021	0.021	0.021
DiffusionSfM	0.025	0.025	0.025	0.025	0.026	<u>0.025</u>	0.028
RD*+MoGe [41]	0.071	0.075	0.068	0.067	0.066	0.064	0.064
DUST3R* [36]	0.038	0.036	0.036	0.036	0.034	0.033	0.034
DUST3R [36]	0.023	0.022	0.019	0.020	0.019	0.020	0.020
DiffusionSfM*	0.028	0.025	0.024	0.024	0.024	0.023	0.022
DiffusionSfM	<u>0.027</u>	<u>0.024</u>	<u>0.023</u>	<u>0.022</u>	<u>0.022</u>	<u>0.022</u>	<u>0.021</u>

Table 2. **Chamfer Distance ($\downarrow$) on CO3D Unseen Categories.** **Top:** CD computed on all scene points. **Bottom:** CD computed on foreground points only. Models marked with “*” are trained on CO3D only, while those without are trained on multiple datasets. Note that top and bottom values are not directly comparable, as each ground-truth point cloud is individually normalized.

4.4. Inference Speed

Though our method requires iterative diffusion denoising at inference, we can speed this up by performing early stopping. Specifically, we can treat the x_0 -prediction from early timesteps as our output instead of iterating over all denoising timesteps. Consistent with observations by Zhang *et al.* [41], this in fact yields more accurate predictions than the final-step diffusion outputs. As a result we only require 10 denoising diffusion timesteps for inference, taking 1.91

Hab.	DUST3R	97.0/100	95.0/97.6	94.3/95.0	94.2/93.1
	DiffusionSfM	92.6/100	93.7/ 99.1	94.2/ 98.8	94.9/98.6
RE10	DUST3R	98.1/100	97.7/68.7	97.6/57.9	97.7/53.3
	DiffusionSfM	97.8/100	97.5/ 75.6	97.7/66.2	97.9/61.9

Table 3. **Camera Rotation and Center Accuracy on Two Scene-Level Datasets.** **Top:** Habitat (2–5 views). **Bottom:** RealEstate10k (2, 4, 6, 8 views). Each grid reports camera rotation accuracy (left, $\uparrow$) and center accuracy (right, $\uparrow$). While DiffusionSfM performs on par with DUST3R in rotation accuracy, it consistently surpasses DUST3R in center accuracy.

# of Images	2	3	4	5	6	7	8
DiffusionSfM*	0.019	0.021	0.022	0.022	0.021	0.021	0.021
w/o Mask	0.020	0.022	0.023	0.023	0.022	0.022	0.024

Table 4. **Ablation Study on GT Mask Conditioning for CO3D Unseen Categories.** We assess the effect of replacing GT mask conditioning with depth interpolation in our CO3D variant (DiffusionSfM*), by reporting the CD for predicted geometry. Incorporating mask conditioning to indicate missing data during training improves geometry quality.

seconds on a single A5000 GPU with 8 input images. In contrast, DUST3R takes 8.73 seconds to run the complete pairwise inference and global alignment procedure. We provide additional analysis in the supplementary material.

4.5. Ablation Study

Homogeneous Coordinates. The use of homogeneous coordinates for ray origins and endpoints is crucial for stable model training. To assess its impact, we replace the proposed homogeneous representation with standard 3D coordinates in $\mathbb{R}^3$. Our experiments show that this variant is difficult to train and fails to converge. Further details on this experiment are provided in the supplementary material.

GT Mask Conditioning. To evaluate the effectiveness of the proposed GT mask conditioning, we train a baseline model without using this strategy. For missing values in the ground-truth depth maps, we use nearest-neighbor interpolation to fill in invalid pixels. This experiment is conducted on the CO3D dataset, with results presented in Tab. 4. The findings show that removing GT mask conditioning consistently degrades predicted geometry across varying numbers of input views, even when the loss is still masked. While interpolation can effectively fill missing depth within object regions, it often introduces substantial noise in the background (*e.g.*, the sky). This noise negatively impacts diffusion model training, as the entire ray origin and endpoint maps are used as input. We anticipate an even larger performance gap on non-object-centric datasets with more outdoor scenes, such as MegaDepth [11].

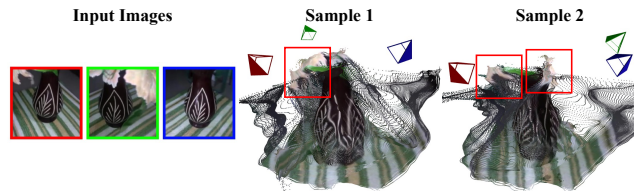


Figure 5. **Multi-modality of DiffusionSfM.** We show two distinct samples from DiffusionSfM, starting from the same input images but with different random noise. Sample 1 explains the input images by putting all flowers on the left side, while Sample 2 places one flower on each side (note the difference in the green camera’s viewpoint). DiffusionSfM is able to predict multi-modal geometry distributions when the scene layout is ambiguous in the inputs.

4.6. Multi-modality from Multiple Sampling

Diffusion models enable the generation of diverse samples from challenging input images. For instance, in Fig. 5, the vase exhibits symmetric patterns, and we present two distinct predicted endpoints from DiffusionSfM, each offering a different interpretation of the flowers in the image. Compared to regression models such as DUST3R [8], DiffusionSfM is better suited for handling uncertainty – an inherent aspect of our task – where a sparse set of input images can correspond to multiple plausible 3D scene geometry.

5. Discussion

We present DiffusionSfM and demonstrate that it recovers accurate predictions of both cameras and geometry from multi-view inputs. Although our results are promising, several challenges and open questions remain.

Notably, DiffusionSfM employs a pixel-space diffusion model, in contrast to the latent-space models adopted by state-of-the-art T2I generative systems. Operating in pixel space may require greater model capacity, yet our current model remains relatively small – potentially explaining the noisy patterns observed along object boundaries. Learning an expressive latent space for ray origins and endpoints by training a VAE could be a promising direction for future work. In addition, the compute requirement in multi-view transformers scales quadratically with the number of input images: one would require masked attention to deploy systems like ours for a large set of input images.

Despite these challenges, we believe that our work highlights the potential of a unified approach for multi-view geometry tasks. We envision that our approach can be built upon to train a common system across related geometric tasks, such as SfM (input images with unknown origins and endpoints), registration (some images have known origins and endpoints, whereas others don’t), mapping (known rays but unknown endpoints), and view synthesis (unknown pixel values for known rays).

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References

- [1] Gilad Baruch, Zhuoyuan Chen, Afshin Dehghan, Tal Dimry, Yuri Feigin, Peter Fu, Thomas Gebauer, Brandon Joffe, Daniel Kurz, Arik Schwartz, et al. Arkitscenes: A diverse real-world dataset for 3d indoor scene understanding using mobile rgb-d data. In *NeurIPS D&B*, 2021. 5
- [2] Herbert Bay, Andreas Ess, Tinne Tuytelaars, and Luc Van Gool. Surf: Speeded up robust features. In *ECCV*, 2006. 2
- [3] Eric Brachmann, Jamie Wynn, Shuai Chen, Tommaso Cavallari, Áron Monszpart, Daniyar Turmukhambetov, and Victor Adrian Prisacariu. Scene coordinate reconstruction: Pos-ing of image collections via incremental learning of a relocalizer. In *ECCV*, 2024. 2
- [4] Shawn T Brown, Paola Buitrago, Edward Hanna, Sergiu Sanielevici, Robin Scibek, and Nicholas A Nystrom. Bridges-2: A platform for rapidly-evolving and data intensive research. In *Practice and experience in advanced research computing*, 2021. 9
- [5] Ruojin Cai, Joseph Tung, Qianqian Wang, Hadar Averbuch-Elor, Bharath Hariharan, and Noah Snavely. Doppelgangers: Learning to disambiguate images of similar structures. In *ICCV*, 2023. 1
- [6] Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superpoint: Self-supervised interest point detection and description. In *CVPR*, 2018. 1, 2
- [7] Bardienus Duisterhof, Lojze Zust, Philippe Weinzaepfel, Vincent Leroy, Yohann Cabon, and Jerome Revaud. Mast3r-sfm: a fully-integrated solution for unconstrained structure-from-motion. In *3DV*, 2025. 2
- [8] Johan Edstedt, Qiyu Sun, Georg Bökman, Mårten Wadenbäck, and Michael Felsberg. RoMa: Robust Dense Feature Matching. In *CVPR*, 2024. 2, 8
- [9] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020. 2, 3
- [10] Vincent Leroy, Yohann Cabon, and Jérôme Revaud. Grounding image matching in 3d with mast3r, 2024. *ECCV*. 1, 2
- [11] Zhengqi Li and Noah Snavely. Megadepth: Learning single-view depth prediction from internet photos. In *CVPR*, 2018. 2, 4, 5, 8
- [12] Amy Lin, Jason Y Zhang, Deva Ramanan, and Shubham Tulsiani. Relpose++: Recovering 6d poses from sparse-view observations. In *3DV*, 2024. 2, 4, 6, 7
- [13] Philipp Lindenberger, Paul-Edouard Sarlin, Viktor Larsson, and Marc Pollefeys. Pixel-perfect structure-from-motion with featuremetric refinement. In *ICCV*, 2021. 2
- [14] Philipp Lindenberger, Paul-Edouard Sarlin, Viktor Larsson, and Marc Pollefeys. Pixel-Perfect Structure-from-Motion with Featuremetric Refinement. In *ICCV*, 2021. 1
- [15] Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. Lightglue: Local feature matching at light speed. In *ICCV*, 2023. 2
- [16] David G Lowe. Distinctive image features from scale-invariant keypoints. In *IJCV*, 2004. 2
- [17] Nikolaus Mayer, Eddy Ilg, Philip Hausser, Philipp Fischer, Daniel Cremers, Alexey Dosovitskiy, and Thomas Brox. A large dataset to train convolutional networks for disparity, optical flow, and scene flow estimation. In *CVPR*, 2016. 5
- [18] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. In *TMLR*, 2024. 3, 4
- [19] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *ICCV*, 2023. 2, 3, 4
- [20] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *ICCV*, 2021. 3, 4
- [21] Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. Common objects in 3d: Large-scale learning and evaluation of real-life 3d category reconstruction. In *ICCV*, 2021. 2, 4, 5, 7
- [22] Paul-Edouard Sarlin, Cesar Cadena, Roland Siegwart, and Marcin Dymczyk. From coarse to fine: Robust hierarchical localization at large scale. In *CVPR*, 2019. 6, 7
- [23] Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superglue: Learning feature matching with graph neural networks. In *CVPR*, 2020. 1, 2
- [24] Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, et al. Habitat: A platform for embodied ai research. In *ICCV*, 2019. 2, 5, 7
- [25] Johannes L Schonberger and Jan-Michael Frahm. Structure-from-motion revisited. In *CVPR*, 2016. 1, 2
- [26] Samarth Sinha, Jason Y Zhang, Andrea Tagliasacchi, Igor Gilitschenski, and David B Lindell. Sparsepose: Sparse-view camera pose regression and refinement. In *CVPR*, 2023. 2
- [27] Cameron Smith, David Charatan, Ayush Tewari, and Vincent Sitzmann. Flowmap: High-quality camera poses, intrinsics, and depth via gradient descent. In *3DV*, 2025. 2

- [28] Noah Snavely, Steven M Seitz, and Richard Szeliski. Photo tourism: exploring photo collections in 3d. In *ACM SIGGRAPH*, 2006. 2
- [29] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, 2021. 3
- [30] Jiaming Sun, Zehong Shen, Yuang Wang, Hujun Bao, and Xiaowei Zhou. Loftr: Detector-free local feature matching with transformers. In *CVPR*, 2021. 2
- [31] Pei Sun, Henrik Kretzschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James Guo, Yin Zhou, Yuning Chai, Benjamin Caine, et al. Scalability in perception for autonomous driving: Waymo open dataset. In *CVPR*, 2020. 5
- [32] Chengzhou Tang and Ping Tan. Ba-net: Dense bundle adjustment network. In *ICLR*, 2019. 2
- [33] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, 2017. 4
- [34] Jianyuan Wang, Christian Rupprecht, and David Novotny. Posediffusion: Solving pose estimation via diffusion-aided bundle adjustment. In *ICCV*, 2023. 2, 6, 7
- [35] Ruicheng Wang, Sicheng Xu, Cassie Dai, Jianfeng Xiang, Yu Deng, Xin Tong, and Jiaolong Yang. Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision. In *CVPR*, 2025. 6
- [36] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *CVPR*, 2024. 1, 2, 4, 5, 6, 7
- [37] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy, 2024. <https://github.com/naver/dust3r>. 5, 6
- [38] Yao Yao, Zixin Luo, Shiwei Li, Jingyang Zhang, Yufan Ren, Lei Zhou, Tian Fang, and Long Quan. Blendedmvs: A large-scale dataset for generalized multi-view stereo networks. In *CVPR*, 2020. 5
- [39] Chandan Yeshwanth, Yueh-Cheng Liu, Matthias Nießner, and Angela Dai. Scannet++: A high-fidelity dataset of 3d indoor scenes. In *ICCV*, 2023. 5
- [40] Jason Y Zhang, Deva Ramanan, and Shubham Tulsiani. Rel-pose: Predicting probabilistic relative rotation for single objects in the wild. In *ECCV*, 2022. 2
- [41] Jason Y. Zhang, Amy Lin, Moneish Kumar, Tzu-Hsuan Yang, Deva Ramanan, and Shubham Tulsiani. Cameras as rays: Sparse-view pose estimation via ray diffusion. In *ICLR*, 2024. 2, 3, 4, 6, 7
- [42] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: Learning view synthesis using multiplane images. In *ACM SIGGRAPH*, 2018. 2, 6, 7